www.csvtuonline.com www.csvtuonline.com

Printed Pages : 4

337832(37)

BE (8th Semester)

Examination, April - May, 2017

[New Scheme]

Finite Element Methods

Time Allowed : 3 hours | *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Attempt any **one** part from (*b*) and (*c*).

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) Write a short note on use of FEM. [2]

(*b*) Explain minimization of domain residual method. A cantilever beam under uniformly distributed load q_0 is as shown in the Fig.-1 (Page-2) whose governing differential equation is given by

$$EI\frac{d^4v}{dx^4} - q_0 = 0 \text{ with boundary conditions,}$$

$$v(0) = 0, \frac{dv}{dx}(0) = 0 \frac{d^2v}{dx^2}(L) = 0, \frac{d^3v}{dx^3}(L) = 0$$

Find the solution for deflection of beam using minimization of domain residual method. [14]

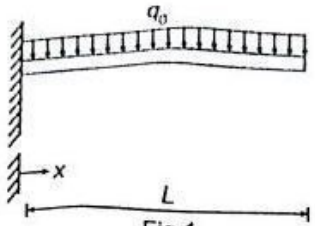


Fig.1

(*c*) Explain weighted residual technique. A uniform bar as shown if Fig. 2 is subjected to linearly distributed load $q = ax$ whose differential equation is given by $AE\frac{d^2u}{dx^2} + ax = 0$ subjected to boundary conditions $u(0) = 0, \frac{du}{dx}(L) = 0$. Find the solution for deformation of bar using Galerkin weighted residual technique. [14]

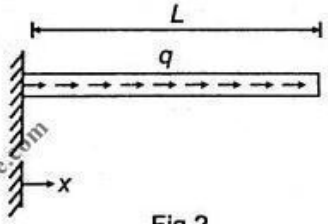


Fig.2

2. (*a*) What are one-dimensional elements? [2]

(*b*) Derive the shape functions for liner bar element. A 1 m long steel rod (3×10 mm cross-section) held fixed at its left end and subjected to concentrated horizontal force of 100 N at its right end. Use two-element mesh (liner element) to find deformation at node points and also find reaction at the fixed support. Assume $E = 2\times10^{11}$ N/m^2. [14]

www.csvtuonline.com www.csvtuonline.com

(*c*) Find temperature at nodes using two equal size elements for a aluminum pin fin having 1 mm diameter, 50 mm long, $k = 200$ W / m / °C, $h = 20$ W / m^2 / °C as shown in Fig. 3. Governing differential equation $k\frac{d^2T}{dx^2} = \frac{Ph}{A_c}(T - T_\infty)$ boundary conditions are insulated tip, T_∞ (ambient temperature) = 30 °C, T_w (wall temperature) = 300°C and no internal heat generation. [14]

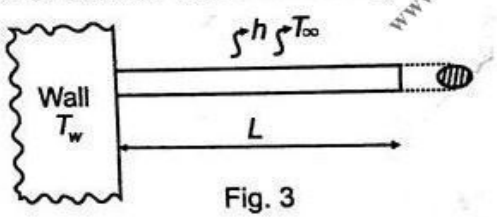
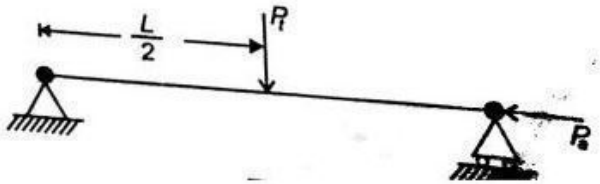


Fig. 3

3. (*a*) How frame element differs from beam element? [2]

(*b*) Derive shape function and stiffness matrix for beam element. [14]

(*c*) Simply supported beam as shown in Fig.4 is subjected to combined axial (P_a) and transverse load (P_t). Use two frame elements to find deflections and slope at nodal points. Assume $EI = 1$, $L = 2$, $AE = 1$, $P_t = 1$ and $P_a = 1$ [14]



www.csvtuonline.com www.csvtuonline.com

4. (*a*) What do you understand by Constant Strain Triangle (CST)? [2]

(*b*) What are natural coordinates? Explain in detail. Derive shape functions for four noded quadrilateral elements in terms of natural coordinates. [14]

(*c*) Evaluate the integral $I = \int_{-1}^{1} \cos\frac{\pi x}{2} dx$ using two-point, three-point and four-point Gauss quadrature rule. [14]

5. (*a*) What do you mean by plane stress and plain strain? [2]

(*b*) Derive strain-displacement matrix [*B*] for four two-dimensional rectangular element. [14]

(*c*) Find out the nodal deflections (u and v) for the thin plate under uniform tension problem shown in Fig. 5.
Assume $E = 2\times10^7$ N/cm^2 and $\vartheta = 0.3$ [14]

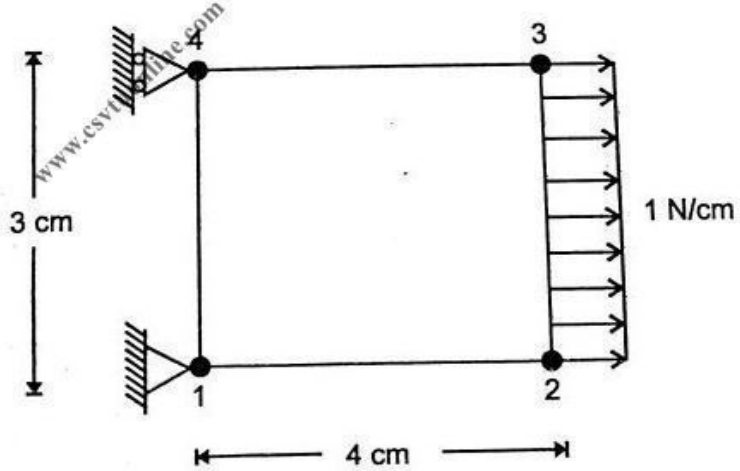


Fig. 5